

Δ	0	0.05	-0.05
Anneal	24.06	24.03	23.88
Static	23.97	23.55	23.63

Table 6: The performance of vanilla NAT with deviation Δ from the optimal setting. ‘Anneal’ represents annealing and ‘Static’ represents the static α .

Models	Vanilla NAT	CMLM	CTC
w/o rephraser	2.07	2.64	0.26
w rephraser	1.35	1.52	0.06

Table 7: The average entropy of NAT models. Lower entropy indicates higher prediction confidence.

measure the prediction confidence, where lower entropy indicates higher confidence. Table 7 shows that NAT models with rephraser have much lower entropy than baseline models, illustrating the effectiveness of rephraser in alleviating the multi-modality problem.

Token Repetitions With low prediction confidence, NAT may generate a mixture of many possible translations, which makes the translation inconsistent and typically contains many repetitive tokens. Therefore, we report the percentage of token repetitions in Table 8, which shows that our approach can significantly reduce token repetitions. Notice that we do not report the results of CTC since it naturally contains no token repetitions.

4.5 Case Study

We present a case of the rephraser output in Table 9 to better understand the working mechanism of rephraser. Before rephrasing, the reference is not aligned with NAT output since there exists the multi-modality problem (“but our loyalty is clear.” and “but it is not about our loyalty.”). After rephrasing, the reference basically keeps the original semantics and is well aligned with NAT output.

5 Related Work

? first proposed non-autoregressive Transformer to reduce the latency of machine translation by generating outputs in parallel. However, there is a large performance gap between the AT and the earliest NAT model, which is mainly attributed to the multi-modality problem.

A number of efforts have explored ways to help NAT handle the multi-modality problem. One thread of research reduces the modality in the target space by leaking part of the target information in the training. ???? augmented NAT with latent variables based on vector quantization or variational inference. Besides, ???? explored the use of word alignments to help the generation of NAT, and ?? enriched NAT with syntactic structures.

Another thread of research focuses on designing robust training objectives for NAT to mitigate the effect of multi-modality. ? introduced regularization terms to reduce errors of repeated and incomplete translations. (?) viewed NAT as

Models	Vanilla NAT	CMLM
w/o rephraser	10.2%	15.4%
w rephraser	2.6%	2.3%

Table 8: The percentage of token repetitions in the generated outputs of Vanilla NAT and CMLM.

Reference	Our identity might sometimes be fu@@ zz@@ y , but our loy@@ alty is clear .
NAT	Our identity may sometimes be urred , but it is not about our loy@@ alty .
Rephraser	Our identity might sometimes be vague , but that is not about our loy@@ alty .

Table 9: A case of the rephraser output from the validation set of WMT14 De-En.

an inference network trained to minimize the autoregressive teacher energy. ??? introduced sequence-level training objectives for NAT. ?? improved the cross-entropy loss with better alignments. Recently, the CTC loss (?) is receiving increasing attention in NAT (????), which is further enhanced with a directed acyclic graph to explicitly model the probability of transistion paths (??).

Our work is closest to diverse distillation (?), which provides multiple references in the dataset and dynamically select one reference to train the model. Diverse distillation has a high distillation cost, and the provided references cannot ensure that all possible outputs are covered. Our work overcomes these limitations by directly rephrasing the reference, which has a lower cost and can provide a more flexible reference.

We train the rephraser with the reinforcement learning technique (?), which is widely used in neural machine translation to optimize sequence-level objectives like BLEU (????). There are also works on paraphrasing the reference to provide a better training target for NMT (???), where the paraphrasing is conducted by human or a paraphrase dictionary.

6 Conclusion

Since the reference may be inappropriate for the training of NAT, we propose rephrasing the reference to provide a better training target for NAT. We apply reinforcement learning to obtain a good rephraser and train NAT based on the rephraser output. Experiments on major benchmarks and NAT baselines demonstrate the effectiveness of our method.

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